THE TWO-COPY WINDOW GRAPH OF A PERMUTIVE SECOND-ORDER CELLULAR AUTOMATON

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ABSTRACT. This note is a supplement to [1]. For a second-order automaton F_f with f a left-permutive elementary rule, the two-copy window graph $\Gamma_{w,M}$ follows both configurations of a pair that differ behind a front moving at the speed of light, on the w cells behind the front and the M cells ahead of it. We determine its eventual image: it has $2^{3w+2}4^M$ states and every walk of length $\lceil w/2 \rceil$ ends in it. For the rules 30, 45, 75 and 135 we prove that it is strongly connected for all w and M. Strong connectivity is reduced to the holonomy of closed walks at each layer behind the front; for 30, 45 and 75 this is read off from explicit periodic histories, and for 135 it is obtained from a single closed walk of odd length by an induction on the depth.

1. INTRODUCTION

The proofs of [1] follow one configuration along the front, through the graphs Λ_M. Here we follow both configurations of a front pair, through a *two-copy* window graph $\Gamma_{w,M}$ that also records w cells behind the front, and we determine its eventual image: its size, the number of steps after which every walk has entered it, and, for the rules 30, 45, 75 and 135, its strong connectivity (Theorem 3.1 and Proposition 8.2). Strong connectivity is reduced to a statement about the holonomy of closed walks at each layer behind the front. For 30, 45 and 75 this holonomy is read off from explicit periodic histories, one family for all depths. For 135 we know no such family, and the holonomy is obtained from a single closed walk of odd length by an induction on the depth (Proposition 7.3). No theorem of [1] uses these results. We use the notation of [1] throughout, and references of the form "Section 3 of [1]" are to that paper.

2. WINDOW STATES

Throughout $f(a,b,c) = a \oplus g(b,c)$ is a left-permutive elementary rule, with $g(b,c) = g_{11}bc \oplus g_{10}b \oplus g_{01}c \oplus g_{00}$ as in Section 5.1 of [1], and $F = F_f$. This is the automaton of Section 4 of [1] with $r = 1$, and we use the diagrams, pairs and front pairs of Section 4.2 of [1]; for rule 30 they are those of Section 3.1 of [1]. Every diagram satisfies

$$Y(t+1,i) \oplus Y(t-1,i) \oplus Y(t,i-1) = g\big(Y(t,i), Y(t,i+1)\big). \tag{1}$$

Fix integers $w \geq 1$ and $M \geq 0$. A *window state* is a pair $s = (a,b)$ of words $a = a_{p-w} \cdots a_{p+M}$ and $b = b_{p-w} \cdots b_{p+M}$ over $\mathcal{A}$, indexed by the sites $p-w, \ldots, p+M$ for some integer p (the *front*), such that

$$a_i = b_i \quad (p < i \leq p+M), \qquad y(a_p) \neq y(b_p), \qquad x(a_p) = x(b_p). \tag{2}$$

States that differ only by a translation of the indices are identified, and $\mathcal{W} = \mathcal{W}_{w,M}$ denotes the finite set of window states. The *window* of a front pair $P = (c, c')$ at p is

$$W(P) = \big(c_{p-w} \cdots c_{p+M},\ c'_{p-w} \cdots c'_{p+M}\big) \in \mathcal{W}.$$

Every $s \in \mathcal{W}$ is the window of some front pair: extend a and b by common cells.

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For $s = (a, b) \in \mathcal{W}$ with front p and $\iota = (\iota_1, \iota_2) \in \mathcal{A}^2$, let $\tilde{a}, \tilde{b}$ be configurations that agree with a, b on $[p - w, p + M]$ and satisfy $\tilde{a}_{p+M+1} = \tilde{b}_{p+M+1} = \iota_1$ and $\tilde{a}_{p+M+2} = \tilde{b}_{p+M+2} = \iota_2$. Define $s \cdot \iota = (a', b')$ as the pair of words $a'_i = F(\tilde{a})_i$, $b'_i = F(\tilde{b})_i$ for $p + 1 - w \le i \le p + 1 + M$, with front $p + 1$. Since F has radius 1, $s \cdot \iota$ does not depend on the other cells of $\tilde{a}, \tilde{b}$. The *window graph* $\Gamma = \Gamma_{w,M}$ has vertex set $\mathcal{W}$ and an edge $s \to s \cdot \iota$ for every $s \in \mathcal{W}$ and $\iota \in \mathcal{A}^2$.

Lemma 2.1. *If $P = (c, c')$ is a front pair at p with $W(P) = s$, then FP is a front pair at $p + 1$ and $W(FP) = s \cdot (c_{p+M+1}, c_{p+M+2})$. In particular $s \cdot \iota \in \mathcal{W}$ for all s, ι. For every $s \in \mathcal{W}$ and $\iota \in \mathcal{A}^2$ there is a front pair P with $W(P) = s$ and $W(FP) = s \cdot \iota$.*

Proof. FP is a front pair at $p + 1$ by Lemma 4.12 of [1](1) with $r = 1$. The cells $F(c)_i$ with $p+1-w \le i \le p+1+M$ read only $c_{p-w}, \dots, c_{p+M+2}$, and $c_i = c'_i$ for $i > p$; this gives the formula for $W(FP)$. For the last statement, extend s by common cells with ι_1, ι_2 at $p+M+1, p+M+2$. □

Lemma 2.2. *Let $s = (a, b) \in \mathcal{W}$ have front p, and let $\iota = (\iota_1, \iota_2) \in \mathcal{A}^2$. Then $s \cdot \iota$ depends on ι only through*

$$(u, v) = \big(y(\iota_1),\ x(\iota_1) \oplus f(y(a_{p+M}), y(\iota_1), y(\iota_2))\big) \in \{0, 1\}^2,$$

and the map $\iota \mapsto (u, v)$ is 4-to-1 onto $\{0, 1\}^2$. The new cell at site $p + M + 1$ is (u, v) in a' and $(u, v \oplus [M = 0])$ in b'. Hence every state has exactly four successors, which are distinct; we write them $s \cdot (u, v)$.

Proof. The new cells at sites $i \le p + M$ read only $\tilde{a}_{i-1}, \tilde{a}_i, \tilde{a}_{i+1}$; site $p + M + 1$ enters only for $i = p + M$, and only through its y-bit u. The new cell at $p + M + 1$ is $F(\tilde{a})_{p+M+1} = (u, v)$. In b the same formula applies with b_{p+M}. If $M \ge 1$, then $b_{p+M} = a_{p+M}$. If $M = 0$, then b_p differs from a_p only in the y-bit, which enters f as its left argument, hence through $\oplus$, so the new cell is $(u, v \oplus 1)$. For each u and each of the four values of ι_2, exactly one value of $x(\iota_1)$ gives a prescribed v, so each (u, v) arises from exactly four ι. Different (u, v) give different new cells at $p + M + 1$, so the four successors are distinct. □

Layer coordinates. For a window state $s = (a, b)$ with front p and each copy $c \in \{a, b\}$ put

$$y_c[d] = y(c_{p-d}) \ \ (-M \le d \le w), \qquad x_c[e] = x(c_{p-e-1}) \ \ (-M - 1 \le e \le w - 1). \tag{3}$$

Thus the cell at site $p - d$ is $(x_c[d - 1], y_c[d])$, and this is a bijection between words in $\mathcal{A}^{w+1+M}$ and assignments of bits to the layers $y[-M], \dots, y[w]$ and $x[-M - 1], \dots, x[w - 1]$. Write $\delta y[d] = y_a[d] \oplus y_b[d]$ and $\delta x[e] = x_a[e] \oplus x_b[e]$. In layer terms, (2) says

$$\delta y[d] = [d = 0] \ \ (-M \le d \le 0), \qquad \delta x[e] = 0 \ \ (-M - 1 \le e \le -1). \tag{4}$$

The x-bit of the front cell is $x[-1]$.

Lemma 2.3 (layer equation)**.** *Let $s' = s \cdot (u, v)$, with layers y'_c, x'_c. For each copy $c \in \{a, b\}$:*

1. *$x'_c[e] = y_c[e]$ for $-M \le e \le w - 1$, and $x'_c[-M - 1] = u$;*
2. *$y'_c[-M] = v_c$, where $v_a = v$ and $v_b = v \oplus [M = 0]$;*
3. *for $-M + 1 \le d \le w$,*

$$y'_c[d] = y_c[d] \oplus x_c[d - 2] \oplus g\big(y_c[d - 1], y_c[d - 2]\big), \tag{5}$$

with the convention $y_c[-M - 1] = u$.

Proof. The new cell at site $i \le p+M$ is $(y(c_i),\ x(c_i) \oplus y(c_{i-1}) \oplus g(y(c_i), y(c_{i+1})))$, with $y(c_{p+M+1}) = u$. With front $p + 1$, $x'_c[e]$ is the x-bit of the new cell at $p - e$, which is $y(c_{p-e}) = y_c[e]$ for $p - e \le p + M$; for $e = -M - 1$ it is the x-bit u of the new cell at $p + M + 1$ (Lemma 2.2). Likewise $y'_c[-M]$ is the y-bit of the new cell at $p + M + 1$. For $-M + 1 \le d \le w$, $y'_c[d]$ is the y-bit of the new cell at $i = p + 1 - d \le p + M$, and $x(c_i) = x_c[d - 2]$, $y(c_{i-1}) = y_c[d]$, $y(c_i) = y_c[d - 1]$, $y(c_{i+1}) = y_c[d - 2]$, where $y_c[-M - 1]$ stands for $y(c_{p+M+1}) = u$. □

If P is a front pair at p and $z_t[d]$ denotes the y-bit of $F^t(c)$ at site $p+t-d$, then the window of F^tP has y-layers z_t and x-layers z_{t-1}, and (5) is (1) written in these comoving coordinates. It is lower triangular: $y'[d]$ reads only layers with index at most d, and the leftmost x-layer $x[w-1]$ is never read.

3. The eventual image

Theorem 3.1. *Let f be rule 30, and let $w \geq 1$ and $M \geq 0$. There is a set $\mathcal{T} = \mathcal{T}_{w,M} \subseteq \mathcal{W}_{w,M}$ with $|\mathcal{T}| = 2^{3w+2}4^M$ such that:*

1. *if $s \in \mathcal{T}$, then $s \cdot \iota \in \mathcal{T}$ for every $\iota \in \mathcal{A}^2$;*
2. *every walk in $\Gamma_{w,M}$ of length at least $\lceil w/2 \rceil$ ends in $\mathcal{T}$;*
3. *the subgraph of $\Gamma_{w,M}$ induced on $\mathcal{T}$ is strongly connected.*

Consequently $\mathcal{T}$ is the unique terminal strongly connected component of $\Gamma_{w,M}$, and for any $s, s' \in \mathcal{W}_{w,M}$ there are walks from s and from s' that end at a common state.

The consequences follow from (1)–(3). Every vertex has out-degree 4, so from s there is a walk of length $\lceil w/2 \rceil$, which ends at some $t \in \mathcal{T}$ by (2). Likewise s' reaches some $t' \in \mathcal{T}$, and by (3) there is a walk from t to t'. So both s and s' reach t'. By (1) and (3), $\mathcal{T}$ is a terminal strongly connected component. Any terminal component T contains the endpoints of walks of length $\lceil w/2 \rceil$ that start in it, so T meets $\mathcal{T}$, and hence $T = \mathcal{T}$.

4. The tower of factor graphs

In this subsection f is any left-permutive elementary rule, except in Lemmas 4.12 and 4.13, which are specific to rule 30. Thus parts (1) and (2) of Theorem 3.1 and the size of $\mathcal{T}$ are proved for every such rule, and part (3) is reduced to the existence of closed walks with prescribed holonomy (Lemma 4.11). Fix $w \geq 1$ and $M \geq 0$, and write $\Gamma = \Gamma_{w,M}$ and $\mathcal{W} = \mathcal{W}_{w,M}$. We use the layer coordinates (3). For a quantity Θ computed from one copy of a state we write Θ^a, Θ^b for its values on the two copies, and $\delta\Theta = \Theta^a \oplus \Theta^b$. A state of $\mathcal{W}$ is called *valid* to stress that it satisfies (4).

Backward rows. For one copy of a window state put $J_0 = [-M, w]$ and $J_m = [-M-1,\ w+1-2m]$ for $m \geq 1$, and define $\beta_m \colon J_m \to \{0,1\}$ by $\beta_0(d) = y[d]$, $\beta_1(e) = x[e]$ and, for $m \geq 1$ and $e \in J_{m+1}$,

$$\beta_{m+1}(e) = \beta_{m-1}(e+2) \oplus \beta_m(e+2) \oplus g\big(\beta_m(e+1), \beta_m(e)\big). \tag{6}$$

The arguments lie in the stated domains. For $e \in J_{m+1}$ we have $e+2 \leq w+1-2m$, so $e, e+1, e+2 \in J_m$. If $m \geq 2$, then $e+2 \in J_{m-1}$. If $m = 1$, then $e+2 \in [-M+1, w-1] \subseteq J_0$. Formula (6) is (5) solved for the earlier row, so β_m recomputes the history m steps back.

Lemma 4.1. *Let $s_0 \to s_1 \to \cdots \to s_n$ be a walk in Γ with $s_{t+1} = s_t \cdot (u_t, v_t)$. For one copy, define $z_t[d]$ for $-1 \leq t \leq n$ as follows:*

- *$z_t[d] = y[d]$ of s_t for $0 \leq t \leq n$ and $-M \leq d \leq w$;*
- *$z_{t-1}[e] = x[e]$ of s_t for $0 \leq t \leq n$ and $-M-1 \leq e \leq w-1$.*

This is consistent, and $z_t[-M-1] = u_t$ for $0 \leq t \leq n-1$. The rows of s_n satisfy $\beta_m(e) = z_{n-m}[e]$ for $0 \leq m \leq n+1$ and $e \in J_m$.

Proof. For $1 \leq t \leq n$ and $-M \leq e \leq w-1$, Lemma 2.3(1) gives $x[e]$ of $s_t = y[e]$ of s_{t-1}, so the two definitions agree; and $x[-M-1]$ of s_{t+1} is u_t. By Lemma 2.3(3), for $0 \leq t \leq n-1$ and $-M+1 \leq d \leq w$,

$$z_{t+1}[d] = z_t[d] \oplus z_{t-1}[d-2] \oplus g\big(z_t[d-1], z_t[d-2]\big),$$

where $z_t[-M-1] = u_t$ is the convention of (5). The cases $m = 0, 1$ of the claim are the definitions. Let $1 \le m \le n$ and $e \in J_{m+1}$, and assume the claim for $m-1$ and m. The displayed relation at $t = n-m \in [0, n-1]$ and $d = e+2 \in [-M+1, w]$, solved for $z_{n-m-1}[e]$, reads $z_{n-m-1}[e] = z_{n-m+1}[e+2] \oplus z_{n-m}[e+2] \oplus g(z_{n-m}[e+1], z_{n-m}[e])$. By induction its right side is the right side of (6). □

The set $\mathcal{T}$. For $0 \le j \le w-1$ define (m_j, e_j) by

$$(m_0, e_0) = (1, 0), \qquad (m_j, e_j) = \left(\tfrac{j+3}{2}, -1\right) \ (j \text{ odd}), \qquad (m_j, e_j) = \left(\tfrac{j+2}{2}, 0\right) \ (j \ge 2 \text{ even}).$$

Then $j = e_j + 2m_j - 2$, and $e_j \in J_{m_j}$ because $e_j \le w + 1 - 2m_j$ is equivalent to $j \le w-1$. The map $j \mapsto (m_j, e_j)$ is a bijection from $[0, w-1]$ onto the set of pairs (m, e) with $m \ge 1$, $e \in \{-1, 0\} \cap J_m$ and $(m, e) \ne (1, -1)$. Condition (C_j) on a window state is

$$(C_j) \qquad \delta\beta_{m_j}(e_j) = [e_j = 0],$$

and $\mathcal{T} = \{s \in \mathcal{W} : \ s \text{ satisfies } (C_j) \text{ for all } 0 \le j \le w-1\}$. By Lemma 4.1, (C_j) says that the history recomputed from the state has its difference where the history of a front pair has it: a difference in the y-bit at the comoving front (layer 0) and none one site ahead of it (layer -1). The excluded pair $(1, -1)$ is the validity condition $\delta x[-1] = 0$.

Lemma 4.2. *For every $e \in J_1$, $\beta_1(e) = x[e]$. For $m \ge 2$, $e \in J_m$ and $j = e + 2m - 2$,*

$$\beta_m(e) = y[j] \oplus x[j] \oplus \Phi_{m,e},$$

where $\Phi_{m,e}$ is a function of $y[i]$ for $e+2 \le i \le j-1$ and of $x[i]$ for $e \le i \le j-1$. In particular $\beta_m(e)$ reads only layers with index in $[e, j]$, and its y-layers have index in $[e+2, j]$.

Proof. Induction on m. For $m = 2$, $\beta_2(e) = y[e+2] \oplus x[e+2] \oplus g(x[e+1], x[e])$. Let $m \ge 2$, $e \in J_{m+1}$ and $j' = e + 2m$. In (6), by induction $\beta_m(e+2) = y[j'] \oplus x[j'] \oplus \Phi_{m,e+2}$, where $\Phi_{m,e+2}$ reads y in $[e+4, j'-1]$ and x in $[e+2, j'-1]$. The term $\beta_{m-1}(e+2)$ is $x[e+2]$ if $m = 2$, and otherwise it reads y in $[e+4, j'-2]$ and x in $[e+2, j'-2]$. The terms $\beta_m(e+1)$ and $\beta_m(e)$ read y in $[e+2, j'-1]$ and x in $[e, j'-1]$. So $\Phi_{m+1,e} = \beta_{m-1}(e+2) \oplus \Phi_{m,e+2} \oplus g(\beta_m(e+1), \beta_m(e))$ has the stated dependence. □

Lemma 4.3. *Let $s \in \mathcal{T}$. Then $\delta\beta_m(e) = [e = 0]$ for all $m \ge 0$ and all $e \in J_m$ with $e \le 0$.*

Proof. Strong induction on m. For $m = 0$ this is validity. For $m = 1$ and $e \le -1$ it is validity, and for $e = 0$ it is (C_0); note $0 \in J_1$ since $w \ge 1$. For $m+1 \ge 2$ and $e \in \{-1, 0\}$, it is (C_j) for the j with $(m_j, e_j) = (m+1, e)$. For $m+1 \ge 2$ and $e \le -2$, (6) gives $\delta\beta_{m+1}(e) = \delta\beta_{m-1}(e+2) \oplus \delta\beta_m(e+2) \oplus \delta g(\beta_m(e+1), \beta_m(e))$. Since $e+2 \le 0$, the first two terms are $[e+2 = 0] \oplus [e+2 = 0] = 0$ by induction. Both arguments of g have $\delta = 0$ by induction, as $e+1 \le -1$, so the last term vanishes. □

Lemma 4.4. (1) *(C_0) is $\delta x[0] = 1$. For $1 \le j \le w-1$, with $\Phi_j = \Phi_{m_j, e_j}$, condition (C_j) is*

$$\delta x[j] = \delta y[j] \oplus \Phi_j^a \oplus \Phi_j^b \oplus [j \text{ even}],$$

and its right side depends only on $y_a[i], y_b[i]$ for $1 \le i \le j$ and on $x_a[i], x_b[i]$ for $-1 \le i \le j-1$. In particular the conditions (C_j) involve only the cells at sites $[p-w, p]$, so $\mathcal{T} = \mathcal{T}_w^\circ \times \mathcal{A}^M$ for a set $\mathcal{T}_w^\circ$ of pairs of words on $[p-w, p]$ that does not depend on M.

(2) *The map $s = (a, b) \mapsto \left(a, (\delta y[d])_{1 \le d \le w}\right)$ is a bijection from $\mathcal{T}$ onto $\mathcal{A}^{w+1+M} \times \{0,1\}^w$. Hence $|\mathcal{T}| = 2^{3w+2} 4^M$.*

Proof. (1) Apply δ to Lemma 4.2; for $j \ge 1$ we have $m_j \ge 2$ and $[e_j = 0] = [j \text{ even}]$. Since $e_j \ge -1$, only layers with index ≥ -1 are read, and y-layers only with index $\ge e_j + 2 \ge 1$. The

layer $x[e]$ with $e \geq -1$ and the layer $y[d]$ with $d \geq 1$ lie in cells at sites $p-e-1$ and $p-d$, which are in $[p-w, p]$. The functions $\Phi_{m,e}$ do not involve M.

(2) Given a and $\delta y[1], \ldots, \delta y[w]$, validity fixes $\delta y[d]$ for $d \leq 0$ and $\delta x[e]$ for $e \leq -1$, and $(C_0), (C_1), \ldots, (C_{w-1})$ determine $\delta x[0], \ldots, \delta x[w-1]$ in turn, because the right side of (C_j) involves only $\delta y[j]$ and layers with index $< j$ of both copies, which are already fixed. So b exists and is unique. Thus $|\mathcal{T}| = 4^{w+1+M} 2^w = 2^{3w+2} 4^M$. □

Entrance, predecessors and closure. Let $\mathrm{Im}_n \subseteq \mathcal{W}$ be the set of endpoints of walks of length n in Γ, so that $\mathrm{Im}_0 = \mathcal{W}$.

Lemma 4.5. *For $n \geq 1$, every $s \in \mathrm{Im}_n$ satisfies (C_j) for all $0 \leq j \leq \min(w-1, 2n-1)$. In particular $\mathrm{Im}_n \subseteq \mathcal{T}$ for $n \geq \lceil w/2 \rceil$.*

Proof. Take a walk $s_0 \to \cdots \to s_n = s$. By Lemma 4.1, $\beta_{m_j}(e_j) = z_{n-m_j}[e_j]$ for both copies whenever $m_j \leq n+1$. If $e_j = 0$ and $m_j \leq n$, then $z_{n-m_j}[0]$ is the layer $y[0]$ of the valid state s_{n-m_j}, so its δ is 1. If $e_j = -1$ and $m_j \leq n+1$, then $z_{n-m_j}[-1]$ is the layer $x[-1]$ of the valid state s_{n-m_j+1}, so its δ is 0. For even j, $m_j \leq n$ is equivalent to $j \leq 2n-2$; for odd j, $m_j \leq n+1$ is equivalent to $j \leq 2n-1$. Finally $w-1 \leq 2n-1$ if and only if $n \geq w/2$. □

Lemma 4.6. *Let $s \in \mathcal{T}$. There are exactly four pairs $(\hat{s}, (u, v))$ with $\hat{s} \in \mathcal{T}$ and $\hat{s} \cdot (u, v) = s$. In particular s has a predecessor in $\mathcal{T}$.*

Proof. Let $\hat{s} \in \mathcal{W}$ have layers $\hat{y}_c, \hat{x}_c$, and let y_c, x_c be the layers of s. By Lemma 2.3, $\hat{s} \cdot (u, v) = s$ if and only if:

(i) $\hat{y}_c[e] = x_c[e]$ for $-M \leq e \leq w-1$, and $u = x_c[-M-1]$ (this value is common to both copies, since $\delta x[-M-1] = 0$);
(ii) $v = y_a[-M]$ (then $y_b[-M] = v \oplus [M = 0]$ holds automatically, since $\delta y[-M] = [M = 0]$);
(iii) for $-M+1 \leq d \leq w$, $\hat{x}_c[d-2] = y_c[d] \oplus \hat{y}_c[d] \oplus g(\hat{y}_c[d-1], \hat{y}_c[d-2])$, with $\hat{y}_c[-M-1] = u$.

So (u, v) is determined by s. For $d \leq w-1$, (i) turns (iii) into $\hat{x}_c[d-2] = y_c[d] \oplus x_c[d] \oplus g(x_c[d-1], x_c[d-2]) = \beta^s_2(d-2)$, where β^s_m are the rows of s; the convention $\hat{y}_c[-M-1] = u = x_c[-M-1]$ is used when $d-2 = -M-1$. This determines $\hat{x}$ on $[-M-1, w-3]$. For $d = w$, (iii) gives $\hat{x}_c[w-2] = y_c[w] \oplus \hat{y}_c[w] \oplus g(x_c[w-1], x_c[w-2])$, where $\hat{y}_c[w]$ is free. Finally $\hat{x}_c[w-1]$ is unconstrained. Hence the candidates are parametrized by $\hat{y}_a[w], \hat{y}_b[w], \hat{x}_a[w-1], \hat{x}_b[w-1]$, and we must decide which give $\hat{s} \in \mathcal{T}$.

Validity of $\hat{s}$. For $-M \leq d \leq 0$, $\delta\hat{y}[d] = \delta x[d]$, which is 0 for $d \leq -1$ by validity of s and 1 for $d = 0$ by (C_0) for s (note $0 \leq w-1$). For $-M-1 \leq e \leq \min(-1, w-3)$, $\delta\hat{x}[e] = \delta\beta^s_2(e) = 0$ by Lemma 4.3. If $w \geq 2$ this covers all $e \leq -1$. If $w = 1$, the remaining condition is $\delta\hat{x}[-1] = 0$, where $-1 = w-2$.

Transfer of conditions. We claim $\beta^{\hat{s}}_0(e) = \beta^s_1(e)$ for $-M \leq e \leq w-1$, and $\beta^{\hat{s}}_m(e) = \beta^s_{m+1}(e)$ for $m \geq 1$ and $e \in J_{m+1}$. For $m = 0$ this is (i), and for $m = 1$ it is the formula for $\hat{x}$ on $J_2 = [-M-1, w-3]$ obtained above. For $m \geq 1$ and $e \in J_{m+2}$, apply (6) to both sides: the arguments are $\beta^{\hat{s}}_{m-1}(e+2)$, $\beta^{\hat{s}}_m(e+2)$, $\beta^{\hat{s}}_m(e+1)$, $\beta^{\hat{s}}_m(e)$ with $e, e+1, e+2 \in J_{m+1} \subseteq J_m$ (and $e+2 \in [-M, w-1]$ when $m-1 = 0$), so by induction they equal $\beta^s_m(e+2)$, $\beta^s_{m+1}(e+2)$, $\beta^s_{m+1}(e+1)$, $\beta^s_{m+1}(e)$, and (6) for s gives $\beta^{\hat{s}}_{m+1}(e) = \beta^s_{m+2}(e)$. Now $(m_j+1, e_j) = (m_{j+2}, e_{j+2})$, and $e_j \in J_{m_j+1}$ if and only if $j \leq w-3$. So for $j \leq w-3$, condition (C_j) for $\hat{s}$ is (C_{j+2}) for s, which holds.

The two remaining conditions. By Lemma 4.4(1) applied to $\hat{s}$, condition (C_j) for $\hat{s}$ reads $\delta\hat{x}[0] = 1$ if $j = 0$, and $\delta\hat{x}[j] = \delta\hat{y}[j] \oplus \hat{\Phi}^a_j \oplus \hat{\Phi}^b_j \oplus [j \text{ even}]$ if $j \geq 1$, where $\hat{\Phi}_j$ reads $\hat{y}[i]$ for $i \leq j-1$ and $\hat{x}[i]$ for $i \leq j-1$.

- $j = w-2$ (when $w \geq 2$). The left side is $\delta\hat{x}[w-2] = \delta y[w] \oplus \delta\hat{y}[w] \oplus \delta g(x[w-1], x[w-2])$, in which only $\delta\hat{y}[w]$ is not yet fixed, and it enters with coefficient 1. The right side

involves $\hat{y}[w-2] = x[w-2]$ and layers of $\hat{s}$ with index at most $w-3$, namely $\hat{y}[i] = x[i]$ and $\hat{x}[i] = \beta_2^s(i)$, all determined by s. So this condition determines $\delta\hat{y}[w]$, and $\hat{y}_a[w]$ is free. For $w = 1$ the same argument applies to the validity condition $\delta\hat{x}[-1] = 0$.

- $j = w-1$. The left side is $\delta\hat{x}[w-1]$. The right side involves $\hat{y}[w-1] = x[w-1]$ and layers of $\hat{s}$ with index at most $w-2$. Among these, $\hat{x}_a[w-2]$ and $\hat{x}_b[w-2]$ depend on the values $\hat{y}_a[w], \hat{y}_b[w]$ chosen in the previous step, and everything else is determined by s; $\hat{x}[w-1]$ does not occur on the right. So, for each choice made in the previous step, this condition determines $\delta\hat{x}[w-1]$, and $\hat{x}_a[w-1]$ is free.

Hence $\hat{s} \in \mathcal{T}$ if and only if $\delta\hat{y}[w]$ and $\delta\hat{x}[w-1]$ take the values just described, which leaves exactly two choices of $\hat{y}_a[w]$ and two of $\hat{x}_a[w-1]$, and the input (u, v) is unique. □

Proposition 4.7. (1) *$\mathcal{T} \subseteq \mathrm{Im}_n$ for every $n \geq 0$.*
(2) *$\mathrm{Im}_n = \mathcal{T}$ for every $n \geq \lceil w/2 \rceil$.*
(3) *If $s \in \mathcal{T}$, then $s \cdot (u, v) \in \mathcal{T}$ for all (u, v). Inside $\mathcal{T}$ every state has out-degree 4 and in-degree 4.*
(4) *Every walk in Γ of length at least $\lceil w/2 \rceil$ ends in $\mathcal{T}$.*

Proof. (1) Induction on n: if $\mathcal{T} \subseteq \mathrm{Im}_n$, then every $s \in \mathcal{T}$ has a predecessor in $\mathcal{T} \subseteq \mathrm{Im}_n$ by Lemma 4.6, so $s \in \mathrm{Im}_{n+1}$. (2) Combine (1) with Lemma 4.5. (3) With $n = \lceil w/2 \rceil$, $s \in \mathrm{Im}_n$ gives $s \cdot (u, v) \in \mathrm{Im}_{n+1} = \mathcal{T}$. The degrees come from Lemmas 2.2 and 4.6. (4) The endpoint of a walk of length n lies in Im_n. □

This proves parts (1) and (2) of Theorem 3.1, and $|\mathcal{T}| = 2^{3w+2}4^M$ is Lemma 4.4(2); so far f is any left-permutive elementary rule.

Strong connectivity. Order the layers as

$$L_1 = x[-M-1],\ L_2 = y[-M],\ L_3 = x[-M],\ L_4 = y[-M+1],\ \ldots,\ x[w-1],\ L_{2(w+1+M)} = y[w],$$

that is, $x[-M-1]$ first and then, for $d = -M, \ldots, w$, the layer $y[d]$ followed by $x[d]$ (when $d \leq w-1$). There are $2(w+1+M)$ coordinates, and each takes values in pairs of bits (the a-bit and the b-bit). Let $P_\nu = \{L_1, \ldots, L_\nu\}$, let π_ν be the restriction of a state to P_ν, and put $S_\nu = \pi_\nu(\mathcal{T})$.

Lemma 4.8. (1) *S_ν is the set of assignments η of bit pairs to P_ν that satisfy those conditions of (4) that involve only coordinates in P_ν, and (C_j) for every j with $x[j] \in P_\nu$.*
(2) *If $L_{\nu+1}$ is an x-layer or a layer $y[d]$ with $d \leq 0$, then each $\eta \in S_\nu$ has exactly two extensions in $S_{\nu+1}$, one for each value of the a-bit of $L_{\nu+1}$. If $L_{\nu+1} = y[d]$ with $d \geq 1$, then each $\eta \in S_\nu$ has exactly four extensions, one for each value of the pair $(y_a[d], y_b[d])$.*
(3) *For $s \in \mathcal{T}$, $\pi_\nu(s \cdot (u, v))$ depends only on $\pi_\nu(s)$ and (u, v).*

Proof. (1) By Lemma 4.4(1), (C_j) has the form $\delta x[j] =$ (a function of $y[i]$, $i \leq j$, and $x[i]$, $i \leq j-1$), and all of these coordinates precede $x[j]$. So every element of S_ν satisfies the stated conditions. Conversely, let η satisfy them, and extend η one coordinate at a time: choose the a-bit arbitrarily; set $y_b[d] = y_a[d] \oplus [d = 0]$ for $d \leq 0$, choose $y_b[d]$ arbitrarily for $d \geq 1$, set $x_b[e] = x_a[e]$ for $e \leq -1$, and choose $x_b[j]$ for $j \geq 0$ so that (C_j) holds, which is possible because (C_j) prescribes $\delta x[j]$ in terms of earlier coordinates. The resulting state is valid and satisfies all (C_j), so it lies in $\mathcal{T}$ and restricts to η. (2) follows from (1) and the same description. (3) By Lemma 2.3, the new value of each coordinate depends only on the input and on coordinates that precede it or equal it: $x'[e]$ reads $y[e]$ or u; $y'[-M]$ reads v; and $y'[d]$ reads $y[d]$, $x[d-2]$, $y[d-1]$, $y[d-2]$ (or u). □

Let Γ_ν be the graph with vertex set S_ν and, for each $\eta \in S_\nu$ and $(u,v) \in \{0,1\}^2$, an edge $\eta \to \eta \cdot (u,v) := \pi_\nu(s \cdot (u,v))$ labelled (u,v), where $s \in \mathcal{T}$ is any state with $\pi_\nu(s) = \eta$. This is well defined by Lemma 4.8(3), and $\eta \cdot (u,v) \in S_\nu$ by Proposition 4.7(3). For $\nu = 2(w+1+M)$, Γ_ν is the subgraph of Γ induced on $\mathcal{T}$. Every walk in Γ_ν lifts to a walk in $\Gamma_{\nu+1}$ with the same labels, starting from any lift of its initial vertex.

Lemma 4.9. *Γ_1 and Γ_2 are complete digraphs with loops; in particular they are strongly connected.*

Proof. $S_1 = \{0,1\}$ is the common value of $x[-M-1]$, and the edge labelled (u,v) goes to u. $S_2 = \{0,1\}^2$ consists of the common value of $x[-M-1]$ and the value of $y_a[-M]$ (and $y_b[-M] = y_a[-M] \oplus [M=0]$). By Lemma 2.3(1),(2), the edge labelled (u,v) goes to (u,v). □

Lemma 4.10. *Let $L_{\nu+1} = x[e]$ with $e \geq -M$. If Γ_ν is strongly connected, so is $\Gamma_{\nu+1}$.*

Proof. Write the vertices of $\Gamma_{\nu+1}$ as (η, ξ) with $\eta \in S_\nu$ and $\xi = x_a[e]$. The layer $x[e]$ is read by Lemma 2.3 only in $y'[e+2]$, which is not in $P_{\nu+1}$, and its new value is $x'[e] = y[e]$ with $y[e] \in P_\nu$. So the edge labelled (u,v) goes from (η,ξ) to $(\eta \cdot (u,v),\, y_a[e](\eta))$. Let $(\eta_1,\xi_1), (\eta_2,\xi_2) \in S_{\nu+1}$. Lift (η_2,ξ_2) to some $s \in \mathcal{T}$. By Lemma 4.6 there are $\hat{s} \in \mathcal{T}$ and (u,v) with $\hat{s} \cdot (u,v) = s$. Put $\hat{\eta} = \pi_\nu(\hat{s})$; then $\hat{\eta} \cdot (u,v) = \eta_2$ and $y_a[e](\hat{\eta}) = x_a[e](s) = \xi_2$ by Lemma 2.3(1). Take a walk from η_1 to $\hat{\eta}$ in Γ_ν and lift it to $\Gamma_{\nu+1}$ starting at (η_1,ξ_1). It ends at $(\hat{\eta},\xi')$ for some ξ', and the edge labelled (u,v) then leads to (η_2,ξ_2). □

Lemma 4.11. *Let $L_{\nu+1} = y[d]$ with $d \geq -M+1$. Let $K = (\mathbb{Z}/2)^2$ and $z = (y_a[d], y_b[d])$ if $d \geq 1$, and let $K = \mathbb{Z}/2$ and $z = y_a[d]$ if $d \leq 0$ (then $y_b[d] = y_a[d] \oplus [d=0]$). For an edge ϵ of Γ_ν from η with label (u,v) put*

$$\varphi_c(\epsilon) = x_c[d-2] \oplus g(y_c[d-1], y_c[d-2]), \tag{7}$$

evaluated on η with $y_c[-M-1] = u$, and $\varphi(\epsilon) = (\varphi_a, \varphi_b) \in K$ if $d \geq 1$, $\varphi(\epsilon) = \varphi_a \in K$ if $d \leq 0$. Then:

1. *the edges of $\Gamma_{\nu+1}$ are $(\eta, z) \to (\eta \cdot (u,v),\, z + \varphi(\epsilon))$, and for $d \leq 0$, $\varphi_a = \varphi_b$;*
2. *for a walk ς in Γ_ν let $\ell(\varsigma) \in K$ be the sum of φ over its edges, and let* Hol *be the subgroup of K generated by the values $\ell(\varsigma)$ of closed walks. If Γ_ν is strongly connected and* Hol $= K$*, then $\Gamma_{\nu+1}$ is strongly connected.*

Proof. (1) is (5). The layers $x[d-2]$, $y[d-1]$, $y[d-2]$ lie in P_ν. For $d \leq 0$ they have index at most -1, so they are common to both copies, as is u; hence $\varphi_a = \varphi_b$, and adding φ preserves the relation $y_b[d] = y_a[d] \oplus [d=0]$. (2) By Lemma 4.8(2), $(\eta, z) \in S_{\nu+1}$ for every $\eta \in S_\nu$ and every $z \in K$. Fix $\eta_2 \in S_\nu$, and let Hol_{η_2} be the set of values $\ell(\varsigma)$ of closed walks ς at η_2. It contains 0 and is closed under addition (concatenation), so it is a submonoid of the finite group K, hence a subgroup. For a closed walk ς at any η, choose walks α from η_2 to η and β from η to η_2. Then $\ell(\varsigma) = \ell(\alpha\varsigma\beta) - \ell(\alpha\beta) \in \mathrm{Hol}_{\eta_2}$, so $\mathrm{Hol}_{\eta_2} = \mathrm{Hol} = K$. Given (η_1, z_1) and (η_2, z_2), take a walk α from η_1 to η_2 and a closed walk ς at η_2 with $\ell(\varsigma) = z_2 - z_1 - \ell(\alpha)$. The lift of $\alpha\varsigma$ starting at (η_1, z_1) ends at (η_2, z_2). □

Lemma 4.12 (stationary states)**.** *Let f be rule 30, so that $g(b,c) = b \vee c$. Let $L_{\nu+1} = y[d]$ with $d \geq -M+1$, so that P_ν consists of the layers up to $x[d-1]$. Let η be an assignment of bit pairs to P_ν that satisfies the conditions of (4) on P_ν, and suppose that for some bit χ and both copies c:*

- *$x_c[i] = y_c[i]$ for $-M \leq i \leq d-1$, and $x_c[-M-1] = \chi$;*
- *with $y_c[-M-1] := \chi$, $y_c[i+1] \leq y_c[i]$ for $-M-1 \leq i \leq d-3$.*

Then $\eta \in S_\nu$, the edge labelled $(\chi, y_a[-M])$ is a loop at η, and on this loop $\varphi_c = y_c[d-1] \wedge \neg y_c[d-2]$.

Proof. The loop. By Lemma 2.3 with $u = \chi$ and $v = y_a[-M]$: $x'[i] = y[i] = x[i]$ for $-M \le i \le d-1$, and $x'[-M-1] = \chi$. Next, $y'_c[-M] = y_a[-M] \oplus [c = b][M = 0] = y_c[-M]$ by validity. For $-M+1 \le i \le d-1$, (5) and $x[i-2] = y[i-2]$ (for $i-2 = -M-1$ both equal $\chi = u$) give

$$y'[i] = y[i] \oplus \big(y[i-2] \oplus (y[i-1] \vee y[i-2])\big) = y[i] \oplus \big(y[i-1] \wedge \neg y[i-2]\big) = y[i],$$

because $i-2 \in [-M-1, d-3]$ and $y[i-1] \le y[i-2]$. The same identity gives φ_c.

Membership. By Lemma 4.8(1) we must check (C_j) for $0 \le j \le d-1$. Let $Z(i) = y_c[i]$ for $-M-1 \le i \le d-1$, so that $y_c = Z$ on $[-M, d-1]$ and $x_c = Z$ on $[-M-1, d-1]$. We claim that $\beta_m(e) = Z(e)$ whenever $e \ge -M-1$ (and $e \ge -M$ if $m = 0$) and $e + \max(2m-2, 0) \le d-1$. For $m = 0, 1$ this holds by assumption. For $m+1 \ge 2$, the arguments $\beta_{m-1}(e+2)$, $\beta_m(e+2)$, $\beta_m(e+1)$ and $\beta_m(e)$ of (6) all satisfy the claim's hypotheses, so $\beta_{m+1}(e) = Z(e+2) \oplus Z(e+2) \oplus (Z(e+1) \vee Z(e)) = Z(e)$, because $e \le d-1-2m \le d-3$ gives $Z(e+1) \le Z(e)$. Every (C_j) with $j \le d-1$ has $e_j + 2m_j - 2 = j \le d-1$, so $\delta\beta_{m_j}(e_j) = \delta Z(e_j)$. For $e_j = 0$ this is $\delta y[0] = 1$. For $e_j = -1$ it is $\delta y[-1] = 0$ if $M \ge 1$, and $\delta\chi = 0$ if $M = 0$. So (C_j) holds. □

Lemma 4.13. *Let f be rule 30. For every $d \in [-M+1, w]$, at the step $L_{\nu+1} = y[d]$ we have* $\mathrm{Hol} = K$.

Proof. In each case we give states satisfying the hypotheses of Lemma 4.12, with $x = y$ on $[-M, d-1]$ in both copies, and read off φ on their loops. Each loop is a closed walk, so its φ lies in Hol.

Case $d \le 0$. All layers in P_ν have index at most -1; take them common, with $\chi = 0$, $y[i] = 0$ for $-M \le i \le d-2$, and $y[d-1] = 1$. Validity holds since there are no differences, and monotonicity concerns only layers up to $d-2$, which are all 0. Then $\varphi = y[d-1] \wedge \neg y[d-2] = 1$ (when $d-2 = -M-1$, $y[d-2] = \chi = 0$), which generates $K = \mathbb{Z}/2$.

Case $d = 1$. Take $\chi = 0$, all common layers ($-M \le i \le -1$) equal to 0, $y_a[0] = \sigma_0$ and $y_b[0] = 1 - \sigma_0$ for a bit σ_0. Then $\delta y[0] = 1$, the x-bit of the front is common, and monotonicity concerns only layers up to -1. Here $\varphi_c = y_c[0] \wedge \neg y_c[-1] = y_c[0]$. So $\sigma_0 = 1$ gives $(1, 0)$ and $\sigma_0 = 0$ gives $(0, 1)$, which generate $(\mathbb{Z}/2)^2$.

Case $d = 2$. Take $\chi = 1$, all common layers equal to 1, $y_a[0] = \sigma_0$, $y_b[0] = 1 - \sigma_0$, and $y_c[1] = \alpha_c$. Monotonicity is $1 \ge \cdots \ge 1 \ge y_c[0]$, which holds. Here $\varphi_c = \alpha_c \wedge \neg y_c[0]$. With $\sigma_0 = 0$, $\alpha_a = 1$, $\alpha_b = 0$ we get $(1, 0)$; with $\sigma_0 = 1$, $\alpha_a = 0$, $\alpha_b = 1$ we get $(0, 1)$.

Case $d \ge 3$. Take $\chi = 1$, all common layers equal to 1, $y_a[0] = 1$, $y_b[0] = 0$, $y_c[i] = 0$ for $1 \le i \le d-2$, and $y_c[d-1] = \alpha_c$. The sequence $\chi, 1, \ldots, 1, y_c[0], 0, \ldots, 0$ up to layer $d-2$ is nonincreasing. Here $\varphi_c = \alpha_c \wedge \neg y_c[d-2] = \alpha_c$, so $(\alpha_a, \alpha_b) = (1, 0)$ and $(0, 1)$ generate $(\mathbb{Z}/2)^2$. □

In the case $d = 2$ of this proof the common layers must equal 1. Monotonicity forces this, because one of $y_a[0], y_b[0]$ is 1. It is also forced by (C_1), which reads $\delta x[1] = \delta y[1] \oplus \neg x[-1]$ (from $\beta_2(-1) = y[1] \oplus x[1] \oplus (x[0] \vee x[-1])$ and $\delta x[0] = 1$), so a state with $x[1] = y[1]$ in both copies must have $x[-1] = 1$.

Proof of Theorem 3.1. Parts (1) and (2) and the size of $\mathcal{T}$ were proved above. For (3), we show by induction on ν that Γ_ν is strongly connected. Lemma 4.9 covers $\nu = 1, 2$. Each later coordinate is either $x[e]$ with $e \ge -M$, covered by Lemma 4.10, or $y[d]$ with $d \ge -M+1$, covered by Lemmas 4.11 and 4.13. For the last value of ν, Γ_ν is the subgraph of Γ induced on $\mathcal{T}$. □

5. Closed walks from periodic histories

For the rules other than 30, the stationary states of Lemma 4.12 are replaced by time-periodic solutions of the layer equation. In this subsection $f = a \oplus g(b, c)$ is a left-permutive elementary rule with $g_{11} = 1$, and $\partial g(b, c) = g(b, c) \oplus g(1 \oplus b, c) = c \oplus g_{10}$. Fix $d \in \mathbb{Z}$ and let $T \ge 1$.

Definition 5.1. A *periodic history below* d of period T is a pair $Z = (Z^a, Z^b)$ of functions $Z^c\colon \mathbb{Z} \times (-\infty, d-1] \to \{0,1\}$, $(t,i) \mapsto Z^c_t(i)$, with $Z^c_{t+T} = Z^c_t$, such that for $c \in \{a,b\}$:

(Z1) $Z^c_{t+1}(i) = Z^c_t(i) \oplus Z^c_{t-1}(i-2) \oplus g\big(Z^c_t(i-1), Z^c_t(i-2)\big)$ for all $t \in \mathbb{Z}$ and $i \le d-1$;

(Z2) $Z^a_t(i) = Z^b_t(i)$ for all t and $i \le -1$, and, if $d \ge 1$, $Z^a_t(0) \ne Z^b_t(0)$ for all t.

If $d \le 0$ the two copies coincide by (Z2), and we write Z as one function. More generally, for any d (in particular $d = 1$), a *single periodic history below* d of period T is one function $Y\colon \mathbb{Z} \times (-\infty, d-1] \to \{0,1\}$ with $Y_{t+T} = Y_t$ that satisfies (Z1); no condition (Z2) is imposed. For $d \le 0$ the single histories below d are exactly the periodic histories below d, with $Z^a = Z^b = Y$. For $d = 1$ a single history is not a periodic history below 1, since (Z2) requires the copies to differ at layer 0; Lemma 5.3(4) turns it into one. For a single history we write φ_t, $\ell(Y)$ and $N(Y,i)$ for the quantities below, computed from Y. For $c \in \{a,b\}$, $t \in \mathbb{Z}$ and $i \le d-1$ put

$$\varphi^c_t = Z^c_{t-1}(d-2) \oplus g\big(Z^c_t(d-1), Z^c_t(d-2)\big), \qquad \ell_c(Z) = \sum_{t=0}^{T-1} \varphi^c_t, \qquad N_c(Z,i) = \sum_{t=0}^{T-1} \big(Z^c_t(i) \oplus g_{10}\big),$$

with sums in $\mathbb{Z}/2$.

Thus (Z1) is (5) for the y-layers of a comoving history, with x-layers equal to the y-layers at the preceding time, as in Lemma 4.1. The bit φ^c_t is the increment that a layer d would have to follow, and $\ell_c(Z)$ is its sum over a period. If Z extends to a periodic history below $d+1$, then $\ell_c(Z) = 0$.

Lemma 5.2. *Let $w \ge 1$, $M \ge 0$ and $-M+1 \le d \le w$, and let Γ_ν be the graph of the tower of Section 4 for f at the step $L_{\nu+1} = y[d]$, so that P_ν consists of the layers $x[-M-1], y[-M], \dots, y[d-1], x[d-1]$. Let Z be a periodic history below d of period T. For $t \in \mathbb{Z}$ let η_t be the assignment to P_ν given by*

$$y_c[i] = Z^c_t(i) \quad (-M \le i \le d-1), \qquad x_c[e] = Z^c_{t-1}(e) \quad (-M-1 \le e \le d-1).$$

Then $\eta_t \in S_\nu$, Γ_ν has an edge $\eta_t \to \eta_{t+1}$ with label $(u_t, v_t) = (Z^a_t(-M-1),\, Z^a_{t+1}(-M))$, and the closed walk $\eta_0 \to \eta_1 \to \cdots \to \eta_T = \eta_0$ has $\ell = (\ell_a(Z), \ell_b(Z))$ if $d \ge 1$ and $\ell = \ell_a(Z)$ if $d \le 0$, in the notation of Lemma 4.11. In particular these values lie in Hol. *Moreover S_ν, the edges of Γ_ν and their increments (7) do not depend on $w \ge d$.*

Proof. Validity. By (Z2), $\delta y[i] = [i = 0]$ for $-M \le i \le \min(0, d-1)$ and $\delta x[e] = 0$ for $e \le -1$. These are the conditions of (4) that involve only P_ν.

Membership. By Lemma 4.8(1) it remains to check (C_j) for $0 \le j \le d-1$. By Lemma 4.2, $\beta_m(e)$ reads only layers with index in $[e, e+2m-2]$, so it is defined on η_n whenever $e + \max(2m-2, 0) \le d-1$; such e lie in J_m because $d \le w$. We claim that for such m, e (with $e \ge -M-1$, and $e \ge -M$ if $m = 0$), the backward rows of η_n satisfy $\beta_m(e) = Z_{n-m}(e)$ in each copy. For $m = 0, 1$ this is the definition of η_n. For $m \ge 1$ the four arguments $(m-1, e+2)$, $(m, e+2)$, $(m, e+1)$ and (m, e) in the recursion for $\beta_{m+1}(e)$ satisfy the same constraint, since $e + 2m \le d-1$. So by induction

$$\beta_{m+1}(e) = Z_{n-m+1}(e+2) \oplus Z_{n-m}(e+2) \oplus g\big(Z_{n-m}(e+1), Z_{n-m}(e)\big) = Z_{n-m-1}(e),$$

the last step being (Z1) at time $n-m$ and layer $e+2 \in [-M+1, d-1]$. Now (C_j) with $j = e_j + 2m_j - 2 \le d-1$ reads $\delta\beta_{m_j}(e_j) = \delta Z_{n-m_j}(e_j) = [e_j = 0]$, by (Z2), since $e_j \in \{-1, 0\}$.

Edges. By Lemma 4.8(3), the edge of Γ_ν from η_t with label (u, v) leads to the assignment computed from η_t and (u, v) by the formulas of Lemma 2.3 restricted to P_ν. With $(u, v) = (u_t, v_t)$, these formulas give $x'[e] = y[e] = Z_t(e)$ and $x'[-M-1] = u_t = Z_t(-M-1)$. They give $y'_c[-M] = v_c = Z^c_{t+1}(-M)$; for $c = b$ and $M = 0$ this uses (Z2), and for $M \ge 1$ the layer is common. For $-M+1 \le i \le d-1$, (5) with the convention $y_c[-M-1] = u_t = Z_t(-M-1)$ gives

$y'_c[i] = Z^c_t(i) \oplus Z^c_{t-1}(i-2) \oplus g(Z^c_t(i-1), Z^c_t(i-2)) = Z^c_{t+1}(i)$ by (Z1). So the edge leads to η_{t+1}. Its increment (7) is $Z^c_{t-1}(d-2) \oplus g(Z^c_t(d-1), Z^c_t(d-2)) = \varphi^c_t$, again with the convention when $d-2 = -M-1$. Summing over a period gives ℓ.

Independence of w. By Lemma 4.8(1) and Lemma 4.4(1), S_ν is cut out by conditions involving only P_ν, and these do not depend on w. The same holds for the formulas of Lemma 2.3 on P_ν. □

Lemma 5.3. *Let Z be a periodic history below d of period T.*

(1) (Swap.) *Exchanging the two copies gives a periodic history below d with (ℓ_a, ℓ_b) exchanged.*
(2) (Top flip.) *Let $\Sigma \subseteq \{a, b\}$ be nonempty, with $\Sigma = \{a, b\}$ if $d - 1 \leq 0$. Complementing $Z^c_t(d-1)$ for all t and all $c \in \Sigma$ gives a periodic history Z' below d with $\ell_c(Z') = \ell_c(Z) \oplus N_c(Z, d-2)$ for $c \in \Sigma$ and $\ell_c(Z') = \ell_c(Z)$ for $c \notin \Sigma$.*
(3) (Restriction.) *For $d' \leq d$, the restriction of Z to layers $i \leq d' - 1$ is a periodic history below d'.*
(4) (Splitting the front.) *If Y is a single periodic history below 1, then (Y, Y'), where Y' equals Y except that $Y'_t(0) = 1 \oplus Y_t(0)$, is a periodic history below 1 with $\ell_a \oplus \ell_b = N(Y, -1)$.*

Proof. (1) and (3) are clear. (2) Among the equations (Z1) for $i \leq d-1$, layer $d-1$ occurs only in the one for $i = d-1$, and only through $Z_{t+1}(d-1) \oplus Z_t(d-1)$, which complementation does not change. Condition (Z2) is preserved: if $d - 1 \leq 0$, both copies are complemented. In φ^c_t, $Z_{t-1}(d-2)$ is unchanged and $g(1 \oplus z, Z_t(d-2)) = g(z, Z_t(d-2)) \oplus \partial g(z, Z_t(d-2)) = g(z, Z_t(d-2)) \oplus Z_t(d-2) \oplus g_{10}$, because $g_{11} = 1$. Summing over t gives the formula. (4) The pair (Y, Y) is not a history below 1, since it violates (Z2), so (2) does not apply to it. But the computation in the proof of (2) is made in one copy at a time and uses only (Z1). Applied with $d = 1$ to the single history Y, it shows that Y' satisfies (Z1) and that $\ell(Y') = \ell(Y) \oplus N(Y, -1)$. So $\ell_b = \ell_a \oplus N(Y, -1)$ for the pair (Y, Y'). Condition (Z2) holds because Y and Y' agree below layer 0 and differ at layer 0. □

Lemma 5.4. *Let $c \in \mathcal{A}^{\mathbb{Z}}$ satisfy $\sigma^n c = c$ and $F^L c = c$, and let $p \in \mathbb{Z}$. Then $Y_t(i) = y\big((F^t c)_{p+t-i}\big)$ is periodic in t with a period dividing $\operatorname{lcm}(n, L)$, satisfies $Y_t(i+n) = Y_t(i)$, and satisfies (Z1) for every $i \in \mathbb{Z}$. So it is a single periodic history below every $d \leq 0$, and below 1.*

Proof. Periodicity is clear. With $s = p + t + 1 - i$, (Z1) at (t, i) reads $Y(t+1, s) = Y(t, s-1) \oplus Y(t-1, s) \oplus g(Y(t, s), Y(t, s+1))$ for the diagram $Y = Y_c$, which is (1). □

We call such a Y the comoving diagram of the *tail* c with *phase* p. At time 0 the front sits at site p, and layer i at time t is site $p + t - i$.

Corollary 5.5. *Let $w \geq d$, and let $M \geq 0$ with $-M + 1 \leq d$. Then* Hol $= K$ *at the step $L_{\nu+1} = y[d]$ in each of the following cases:*

(a) *$d \leq 1$, and there is a single periodic history Y below d with $N(Y, d-2) = 1$;*
(b) *$d \geq 1$, and there is a periodic history Z below d with $\ell_a(Z) \neq \ell_b(Z)$;*
(c) *$d \geq 2$, and there is a periodic history Z below d with $N_a(Z, d-2) \neq N_b(Z, d-2)$.*

If the history is the restriction of a tail or of a history defined on all layers $\leq D - 1$, then the conclusion holds for every such M and w at once, for all $d \leq D$ covered by the case.

Proof. (b) By Lemma 5.2, $(\ell_a, \ell_b) \in \{(1, 0), (0, 1)\}$ lies in Hol, and by Lemma 5.3(1) so does the other element of this set. These two elements generate $K = (\mathbb{Z}/2)^2$. (a) For $d \leq 0$, Y and its top flip (Lemma 5.3(2) with $\Sigma = \{a, b\}$) have ℓ differing by 1, so one of them gives $\ell = 1$, which generates $K = \mathbb{Z}/2$. For $d = 1$, Lemma 5.3(4) gives a history with $\ell_a \neq \ell_b$, and (b) applies. (c) If $\ell_a(Z) \neq \ell_b(Z)$, use (b). Otherwise flip the top layer $d - 1 \geq 1$ of the one copy c with $N_c(Z, d-2) = 1$. By Lemma 5.3(2) this changes ℓ_c and not the other, and (b) applies. The last sentence holds because the history is the same for all M and w. □

6. Histories ahead of the front

For $f = 30$ the loops of Lemma 4.13 are time-constant histories in the sense of Definition 5.1. For 45, 75 and 135 we give histories explicitly. The conventions below are also used in Section 7. Tails are given by their x-row and y-row, with the phase stated by the site at which the front sits at time 0. Every series below is listed over $t = 0, 1, \dots, T-1$. The column φ lists the bits $Z_{t-1}(i-2) \oplus g(Z_t(i-1), Z_t(i-2))$ for $t = 0, \dots, T-1$. Condition (Z1) at layer i says exactly that $Z_{t+1}(i) = Z_t(i) \oplus \varphi_t$, which can be checked row by row. Layers that come from a tail satisfy (Z1) by Lemma 5.4, and need no check.

Ahead of the front and layer 1. For $d \le 1$ we use case (a) of Corollary 5.5 with a tail whose $N(Y, i)$ is odd for every i. Any phase may be used.

- $f = 45$, $g_{10} = 0$. The tail is $x = y = \cdots 111 \cdots$, the configuration $(1,1)^{\mathbb{Z}}$. It is fixed by F, since $f(1,1,1) = 0$. So $Y \equiv 1$, $T = 1$, and $N(Y, i) = 1$.
- $f = 75$, $g_{10} = 1$. The tail is $x = y = \cdots 0101 \cdots$, with $x_s = y_s = [s \text{ odd}]$. It is fixed by F: for even s, $y'_s = x_s \oplus y_{s-1} \oplus g(y_s, y_{s+1}) = 0 \oplus 1 \oplus g(0,1) = 0$, and for odd s, $y'_s = 1 \oplus 0 \oplus g(1,0) = 1$. So $Y_t(i) = [p + t - i \text{ odd}]$, $T = 2$, and $N(Y, i) = (Y_0(i) \oplus 1) + (Y_1(i) \oplus 1) = 1$.
- $f = 135$, $g_{10} = 0$. The tail is $x = y = \cdots 0001\,0001 \cdots$, with $x_s = y_s = [s \equiv 3 \bmod 4]$. Its F-orbit has length 8:

  $$(0001;0001) \to (0001;0110) \to (0110;1001) \to (1001;0100)$$
  $$\to (0100;0100) \to (0100;1001) \to (1001;0110) \to (0110;0001) \to (0001;0001),$$

  written over the sites $0, 1, 2, 3$. With phase $p = 0$ the comoving period is $T = 8$, and the layers $i \equiv -1, -2, -3, -4 \pmod 4$ have series 01101000, 00110100, 10000110, 01000011. Each contains three 1's, so $N(Y, i) = 1$ for all i. The same holds for the other three phases.

(For $f = 30$ the tail $(0,0)^{\mathbb{Z}}$, with $Y \equiv 0$ and $g_{10} = 1$, serves the same purpose.)

Proposition 6.1. *Let $f \in \{30, 45, 75, 135\}$, $w \ge 1$ and $M \ge 0$. Then* Hol $= K$ *at every step $L_{\nu+1} = y[d]$ with $-M + 1 \le d \le \min(1, w)$.*

Proof. For 30 this is the cases $d \le 0$ and $d = 1$ of Lemma 4.13. For 45, 75 and 135, the tails above give, by Lemma 5.4, single periodic histories below every $d \le 1$ with $N(Y, d-2) = 1$, so Corollary 5.5(a) applies. Each tail is defined on all layers, so it serves every M and w at once. □

7. Histories behind the front

The steps $y[d]$ with $d \ge 2$ are behind the front. We prove that they have full holonomy for 45, 75 and 135 (Proposition 7.5). For 45 and 75 the proof uses explicit histories, which are the time-periodic replacements for the stationary states of Lemma 4.12; for 135 it uses one explicit history and Proposition 7.3. The conventions are those of Section 6.

Rule 75 behind the front: a stationary wall. Let $d \ge 2$, and take $T = 1$, so that (Z1) reads $Z(i-2) = g(Z(i-1), Z(i-2))$ for $i \le d-1$. Here $g = \neg b \vee c$, so $g(b,1) = 1$ and $g(b,0) = \neg b$. Put

$$Z(i) = 1 \quad (i \le -1), \qquad Z^a(i) = [i \text{ even}], \quad Z^b(i) = [i \text{ odd}] \quad (0 \le i \le d-2),$$

and $Z^a(d-1) = Z^b(d-1) = 0$. So $Z^a = (\dots 1,1 \mid 1 \mid 0,1,0,\dots)$ and $Z^b = (\dots 1,1 \mid 0 \mid 1,0,1,\dots)$, read from layer -1 through the front at layer 0, with the top layer $d-1$ set to 0. Condition (Z1) asks $Z(k) = g(Z(k+1), Z(k))$ for $k \le d-3$. This holds when $Z(k) = 1$. When $Z(k) = 0$, we have $0 \le k \le d-3$, and the alternating pattern gives $Z(k+1) = 1$ and $g(1,0) = 0$. Condition (Z2) is clear. Now $\varphi^c = Z^c(d-2) \oplus g(0, Z^c(d-2)) = Z^c(d-2) \oplus 1$, and $Z^a(d-2) \ne Z^b(d-2)$, so $\ell_a \ne \ell_b$. Corollary 5.5(b) applies for every $d \ge 2$.

Rule 45 at $d = 2$: a loop of period 4. Here $g = b \vee \neg c$. The tail is $x = y = \cdots 0101 \cdots$, with $x_s = y_s = [s \text{ odd}]$, and phase $p = 0$, so the front sits at a site where the cell is $(0, 0)$. Its F-orbit is the 4-cycle O of Lemma 5.5 of [1], and the comoving period is $T = 4$. For $i \le -1$ the layer series is 0011 for even i and 1100 for odd i. The history below 2 is

layer i	$Z^a(i)$	φ^a	$Z^b(i)$	φ^b
-2	0011	0101	0011	0101
-1	1100	0101	1100	0101
0	0011	0101	1100	0101
1	0011	0101	0111	1001
step $d = 2$		0110		0001

Copy a is the tail on layers ≤ 0. Its layer 1 is the complement of the tail's (1100), so it is a top flip. Copy b is the tail with layer 0 complemented (Lemma 5.3(4)). Only layer 1 of copy b needs checking. There, $\varphi^b_t = Z_{t-1}(-1) \oplus g(Z^b_t(0), Z_t(-1))$ equals $0 \oplus g(1, 1)$, $1 \oplus g(1, 1)$, $1 \oplus g(0, 0)$, $0 \oplus g(0, 0)$, that is, $1, 0, 0, 1$. And 0111 does follow these increments. The last row gives $\ell = (0, 1)$, so Corollary 5.5(b) applies at $d = 2$.

Rule 45 at $d \ge 3$: a time-periodic wall. Take $T = 2$ and the all-ones tail on layers ≤ -1. Since $Z_{t-1} = Z_{t+1}$, condition (Z1) at layer i says that both entries of $\varphi(i) = \big(Z_1(i-2) \oplus g(Z_0(i-1), Z_0(i-2)),\ Z_0(i-2) \oplus g(Z_1(i-1), Z_1(i-2))\big)$ equal $Z_0(i) \oplus Z_1(i)$. The series $(Z_0(i), Z_1(i))$ are:

layer i	≤ -1	0	1	2	3	4	5	6	7	$\cdots$
$Z^a(i)$	11	11	11	00	01	10	00	00	01	period 4 from $i = 2$
$\varphi^a(i)$	00	00	00	00	11	11	00	00	11	
$Z^b(i)$	11	00	01	10	00	00	01	10	00	period 4 from $i = 0$
$\varphi^b(i)$	00	00	11	11	00	00	11	11	00	

For example, $\varphi^a(3) = (Z_1(1) \oplus g(Z_0(2), Z_0(1)),\ Z_0(1) \oplus g(Z_1(2), Z_1(1))) = (1 \oplus g(0, 1),\ 1 \oplus g(0, 1)) = (1, 1)$, which matches $Z^a_0(3) \oplus Z^a_1(3) = 1$. Condition (Z1) at layer i reads only layers $i-2, i-1, i$. By the stated periodicities, these three layers coincide with the three layers four lower when $i - 6 \ge 2$ in copy a and when $i - 6 \ge 0$ in copy b. So the checks at $i \ge 8$ (copy a) and $i \ge 6$ (copy b) repeat checks four layers lower, and every other check, at $i \le 7$ and $i \le 5$ respectively, is visible in the table. (The triples differ at $i = 6, 7$ in copy a and at $i = 5$ in copy b, so these indices cannot be lowered.) Condition (Z2) is clear. With $g_{10} = 0$, $N_c(Z, i) = Z^c_0(i) \oplus Z^c_1(i)$. For $i \ge 1$ this is 1 for copy a exactly when $i \equiv 3, 0 \pmod 4$, and 1 for copy b exactly when $i \equiv 1, 2 \pmod 4$. So $N_a(Z, i) \ne N_b(Z, i)$ for every $i \ge 1$. For every $d \ge 3$, the restriction of Z to layers $\le d - 1$ (Lemma 5.3(3)) satisfies Corollary 5.5(c). At $d = 2$ it does not, since $N_a(Z, 0) = N_b(Z, 0) = 0$; that step is covered by the period-4 loop above.

Rule 135 at $d = 2, 3$. Here $g = \neg(b \wedge c)$. The tail is $x = y = \cdots 011\,011 \cdots$, with $x_s = y_s = [s \not\equiv 0 \bmod 3]$, and phase $p = 0$, so the front sits at a site where the cell is $(0, 0)$. The tail is fixed by F. For $s \equiv 0, 1, 2 \pmod 3$, $y'_s = x_s \oplus y_{s-1} \oplus g(y_s, y_{s+1})$ equals $0 \oplus 1 \oplus g(0, 1) = 0$, $1 \oplus 0 \oplus g(1, 1) = 1$ and $1 \oplus 1 \oplus g(1, 0) = 1$. Its comoving period is $T = 3$, and the layers $i \equiv 0, -1, -2 \pmod 3$ have series 011, 110, 101. Two histories, below 2 and below 3:

	below 2				below 3			
layer i	Z^a	φ^a	Z^b	φ^b	Z^a	φ^a	Z^b	φ^b
-2	101	110	101	110	101	110	101	110
-1	110	011	110	011	110	011	110	011
0	011	101	100	101	011	101	100	101
1	010	110	111	000	010	110	000	000
2					000	000	011	101
step d		000		001		110		111

In both histories, copy a is the tail on layers ≤ 0, and its layer 1 is a top flip of the tail's (101). Copy b is the tail with layer 0 complemented. The remaining rows are checked against the φ columns. For example, for copy b at layer 1, $\varphi^b_t = Z_{t-1}(-1) \oplus g(Z^b_t(0), Z_t(-1))$ equals $0 \oplus g(1,1)$, $1 \oplus g(0,1)$, $1 \oplus g(0,0)$, that is, $0,0,0$, so layer 1 is constant (111 or 000). The step rows give $\ell = (0,1)$ in both cases, so Corollary 5.5(b) applies at $d = 2$ and $d = 3$. Both histories have the odd period 3; the history below 2 supplies hypothesis (ii) of Proposition 7.3 for 135. The same works for the other two phases of this tail.

Remark 7.1 (stationary fronts). Call a periodic history *stationary* if $T = 1$. A stationary history below d with $d \ge 2$ can exist only if $g_{10}(1 + g_{01}) + g_{00} = 0$. Indeed, (Z1) at $i = 1$ reads $Z(-1) = g(Z^c(0), Z(-1))$ for both copies, and $Z^a(0) \ne Z^b(0)$. So $\partial g(\cdot, Z(-1)) = Z(-1) \oplus g_{10}$ vanishes, that is, $Z(-1) = g_{10}$. Then $g_{10} = g(0, g_{10}) = g_{01}g_{10} \oplus g_{00}$. The quantity $g_{10}(1+g_{01})+g_{00}$ is invariant under κ. It is 0 for the classes of 30 and 75 and 1 for those of 45 and 135, so for the latter two classes the histories behind the front must be time-periodic. For 30 and 75, stationary histories for every $d \ge 2$ are given explicitly (Lemma 4.13 and the wall above). We do not claim the converse in general.

All depths from one closed walk of odd length. The histories above reach a fixed depth, or every depth when there is a wall. The next proposition needs neither: one closed walk of odd length at the step $y[2]$ gives full holonomy at every step behind it. Fix a left-permutive rule $f = a \oplus g(b,c)$ with $g_{11} = 1$, and integers $w \ge 2$ and $M \ge 0$. For $1 \le d \le w$ let G_d be the graph Γ_ν of the tower of Section 4 with $L_{\nu+1} = y[d]$. Its vertices are the elements of S_ν, which assign bit pairs to the layers $x[-M-1], y[-M], \dots, y[d-1], x[d-1]$. For $d \ge 2$ these are the layers of G_{d-1} together with $y[d-1]$ and $x[d-1]$. Restricting the vertices of a walk ς in G_d to the layers of G_{d-1} gives a walk $\bar\varsigma$ in G_{d-1} with the same labels (Lemma 4.8(3)), the *projection* of ς. For an edge ϵ of G_d from η put

$$\psi_d(\epsilon) = \big(1,\ \varphi_a(\epsilon),\ \varphi_b(\epsilon),\ y_a[d-1](\eta) \oplus g_{10},\ y_b[d-1](\eta) \oplus g_{10}\big) \in (\mathbb{Z}/2)^5,$$

where $\varphi = (\varphi_a, \varphi_b)$ is the increment (7) at the step $y[d]$, as in Lemma 4.11. For a walk ς in G_d let $\Psi_d(\varsigma)$ be the sum of ψ_d over its edges. We write

$$\Psi_d(\varsigma) = \big(T(\varsigma),\ \ell_a(\varsigma),\ \ell_b(\varsigma),\ N_a(\varsigma),\ N_b(\varsigma)\big),$$

so that $T(\varsigma)$ is the length of ς modulo 2, $\ell(\varsigma) = (\ell_a, \ell_b)$ is the quantity of Lemma 4.11, and $N_c(\varsigma)$ counts, modulo 2, the vertices of ς, the last one excepted, at which $y_c[d-1] \ne g_{10}$. Let $I_d \subseteq (\mathbb{Z}/2)^5$ be the set of values $\Psi_d(\varsigma)$ of closed walks ς in G_d. If ς is the closed walk of a periodic history Z below d of period T (Lemma 5.2), then $\Psi_d(\varsigma) = \big(T, \ell_a(Z), \ell_b(Z), N_a(Z, d-1), N_b(Z, d-1)\big)$.

Lemma 7.2. *Let $2 \le d \le w$.*

(1) *If G_d is strongly connected, then I_d is a subgroup of $(\mathbb{Z}/2)^5$.*
(2) (Top flip.) *For $c \in \{a, b\}$ let τ_c complement the two bits $y_c[d-1]$ and $x_c[d-1]$ of an assignment. Then τ_c maps the vertex set of G_d onto itself and maps every edge to an*

edge with the same label. For a closed walk ς in G_d with projection $\bar\varsigma$,

$$\Psi_d(\tau_a\varsigma) = \Psi_d(\varsigma) + \big(0,\, N_a(\bar\varsigma),\, 0,\, T(\varsigma),\, 0\big), \qquad \Psi_d(\tau_b\varsigma) = \Psi_d(\varsigma) + \big(0,\, 0,\, N_b(\bar\varsigma),\, 0,\, T(\varsigma)\big),$$

where $N_c(\bar\varsigma)$ is computed in G_{d-1}.

(3) (Swap.) *The exchange μ of the two copies maps the vertex set of G_d onto itself and edges to edges. It maps a closed walk ς to a closed walk $\mu\varsigma$ with $\Psi_d(\mu\varsigma) = (T, \ell_b, \ell_a, N_b, N_a)(\varsigma)$ and $N_a(\overline{\mu\varsigma}) = N_b(\bar\varsigma)$, $N_b(\overline{\mu\varsigma}) = N_a(\bar\varsigma)$.*

(4) (Lifting.) *Let ω be a closed walk in G_{d-1} with $\ell(\omega) = (0,0)$, where ℓ is taken at the step $y[d-1]$. Then there is a closed walk $\hat\omega$ in G_d with projection ω; in particular $\hat\omega$ has the same length as ω.*

Proof. (1) Ψ_d is additive under concatenation of walks. So the argument in the proof of Lemma 4.11(2) applies with Ψ_d in place of ℓ. The values of closed walks at a fixed vertex η_2 form a submonoid of a finite group, hence a subgroup. For a closed walk ς at any vertex η, choose walks α from η_2 to η and β from η to η_2; then $\Psi_d(\varsigma) = \Psi_d(\alpha\varsigma\beta) - \Psi_d(\alpha\beta)$ lies in that subgroup. Conversely that subgroup is contained in I_d, so I_d is equal to it.

(2) Let $\Gamma_\nu = G_d$. By Lemma 4.8(1), S_ν is defined by the conditions of (4), which involve only layers $y[i]$ with $i \le 0$ and $x[e]$ with $e \le -1$, and by (C_j) for $0 \le j \le d-1$. By Lemmas 4.2 and 4.4(1), (C_0) is $\delta x[0] = 1$, and for $j \ge 1$ condition (C_j) reads $\delta x[j] \oplus \delta y[j] = \Phi^a_j \oplus \Phi^b_j \oplus [j \text{ even}]$, where Φ_j reads only layers of index less than j. Since $d-1 \ge 1$, the layer $d-1$ occurs only in (C_{d-1}), and there only through $\delta x[d-1] \oplus \delta y[d-1]$, which τ_c does not change. So $\tau_c(S_\nu) = S_\nu$. By Lemma 2.3, the layer $x[d-1]$ is read only by $y'[d+1]$, and the layer $y[d-1]$ is read by $x'[d-1] = y[d-1]$, by $y'[d-1]$, in which it is a summand, and by $y'[d]$ and $y'[d+1]$. The layers $y[d]$ and $y[d+1]$ do not belong to G_d. The other layers of G_d are neither changed by τ_c nor computed from the layer $d-1$. Since $\eta \cdot (u,v)$ is computed from η and (u,v) by these formulas (Lemma 4.8(3)), $\tau_c(\eta \cdot (u,v)) = \tau_c(\eta) \cdot (u,v)$ for every vertex η and every label. Now compare ψ_d on an edge from η and on its image. The increment $\varphi_c = x_c[d-2] \oplus g(y_c[d-1], y_c[d-2])$ changes by $\partial g(y_c[d-1], y_c[d-2]) = y_c[d-2] \oplus g_{10}$, because $g_{11} = 1$ (here $d-2 \ge 0 > -M-1$, so the convention $y_c[-M-1] = u$ does not enter); the increment of the other copy does not change; the entry $y_c[d-1] \oplus g_{10}$ changes by 1. Summing over the edges of ς gives the two formulas, since $\sum_t \big(y_c[d-2](\eta_t) \oplus g_{10}\big) = N_c(\bar\varsigma)$, the sum being over the vertices η_t of ς, the last one excepted.

(3) The conditions (4) and (C_j) prescribe values of δ, so they are symmetric in the two copies. The formulas of Lemma 2.3 are the same for both copies, except that $v_b = v \oplus [M = 0]$. So the exchange maps the edge with label (u,v) to the edge with label $(u, v \oplus [M=0])$, and it exchanges the entries of ψ_d and of ψ_{d-1} belonging to the two copies.

(4) Let $G_{d-1} = \Gamma_{\nu'}$, so that $L_{\nu'+1} = y[d-1]$, $L_{\nu'+2} = x[d-1]$ and $G_d = \Gamma_{\nu'+2}$. Recall that a walk lifts to the next graph of the tower, uniquely, from any lift of its initial vertex. Let $\omega = (\eta_0 \to \cdots \to \eta_T = \eta_0)$; we may assume $T \ge 1$. Since $d-1 \ge 1$, every pair (η_0, z) with $z \in (\mathbb{Z}/2)^2$ is a vertex of $\Gamma_{\nu'+1}$ (Lemma 4.8(2)). By Lemma 4.11(1), the lift of ω to $\Gamma_{\nu'+1}$ starting at (η_0, z) ends at $(\eta_0, z + \ell(\omega)) = (\eta_0, z)$, so it is a closed walk ω_1. Lift ω_1 to $\Gamma_{\nu'+2}$, starting at any lift of its initial vertex. The lift ends at a vertex ζ over the same vertex, and by Lemma 2.3(1) the layer $x[d-1]$ of ζ is the layer $y[d-1]$ of the last vertex but one of ω_1. Lift ω_1 once more, starting at ζ. The new lift ends at a vertex over the same vertex with the same layer $x[d-1]$, that is, at ζ. This closed walk is $\hat\omega$. □

Proposition 7.3. *Let $f = a \oplus g(b,c)$ be left-permutive with $g_{11} = 1$, and let $w \ge 2$ and $M \ge 0$. Suppose that*

(i) $\mathrm{Hol} = K$ *at every step $L_{\nu+1} = y[d]$ with $-M+1 \le d \le 1$, and*
(ii) *G_2 has a closed walk of odd length.*

Then for every $2 \le d \le w$ the graph G_d is strongly connected and $I_d = (\mathbb{Z}/2)^5$. In particular $\mathrm{Hol} = K$ *at every step $L_{\nu+1} = y[d]$ with $2 \le d \le w$.*

Proof. The last statement follows from the first, since $I_d = (\mathbb{Z}/2)^5$ contains elements with every value of (ℓ_a, ℓ_b). We prove the first by induction on d.

The case $d = 2$. By Lemmas 4.9, 4.10 and 4.11(2) and hypothesis (i), every graph of the tower up to G_2 is strongly connected, as in the proof of Theorem 3.1. So I_2 is a group by Lemma 7.2(1). Let ς be a closed walk of odd length in G_2, and put $n_c = N_c(\bar{\varsigma})$, computed in G_1. The top y-layer of G_1 is $y[0]$, and $y_b[0] = y_a[0] \oplus 1$ at every vertex, so $n_a + n_b = T(\varsigma) = 1$. By Lemma 7.2(2), I_2 contains $\Psi_2(\tau_a\varsigma) - \Psi_2(\varsigma) = (0, n_a, 0, 1, 0)$ and $\Psi_2(\tau_b\varsigma) - \Psi_2(\varsigma) = (0, 0, n_b, 0, 1)$. The same applied to $\mu\varsigma$ (Lemma 7.2(3)) gives $(0, n_b, 0, 1, 0)$ and $(0, 0, n_a, 0, 1)$. Since $\{n_a, n_b\} = \{0, 1\}$, these four elements are

$$(0,1,0,1,0), \qquad (0,0,0,1,0), \qquad (0,0,1,0,1), \qquad (0,0,0,0,1). \tag{8}$$

They span $\{0\} \times (\mathbb{Z}/2)^4$. Together with $\Psi_2(\varsigma)$, whose first entry is 1, they generate $(\mathbb{Z}/2)^5$.

The step from $d-1$ to d, for $3 \le d \le w$. Assume that G_{d-1} is strongly connected and $I_{d-1} = (\mathbb{Z}/2)^5$. Then $\mathrm{Hol} = K$ at the step $y[d-1]$, so G_d is strongly connected by Lemmas 4.11(2) and 4.10, and I_d is a group. Choose closed walks ω and ω' in G_{d-1} with

$$\Psi_{d-1}(\omega) = (1,0,0,1,1), \qquad \Psi_{d-1}(\omega') = (1,0,0,0,0).$$

Both have $\ell = (0, 0)$, so they have lifts $\hat{\omega}, \hat{\omega}'$ to closed walks in G_d (Lemma 7.2(4)), of odd length. By Lemma 7.2(2), I_d contains

$$\Psi_d(\tau_a\hat{\omega}) - \Psi_d(\hat{\omega}) = (0,1,0,1,0), \qquad \Psi_d(\tau_b\hat{\omega}) - \Psi_d(\hat{\omega}) = (0,0,1,0,1),$$

because $N_a(\omega) = N_b(\omega) = 1$, and likewise $(0,0,0,1,0)$ and $(0,0,0,0,1)$ from $\hat{\omega}'$, because $N_a(\omega') = N_b(\omega') = 0$. These are the elements (8). Together with $\Psi_d(\hat{\omega})$, whose first entry is 1, they generate $(\mathbb{Z}/2)^5$. □

Remark 7.4. (1) Hypothesis (ii) enters through the entry T: on a closed walk of even length the top flip does not change N_c. It holds as soon as there is a periodic history below 2 of odd period, by Lemma 5.2; such a history serves every M and every $w \ge 2$. For 30 and 75 the stationary histories below 2 (the case $d = 2$ of Lemma 4.13, and the wall above) have period 1, so Proposition 7.3 gives a second proof of the case $d \ge 2$ for these rules.

(2) The proof uses $g_{11} = 1$ only in the top flip. For an affine rule, $\partial g = g_{10}$ is constant, the top flip changes Ψ_d by $(0, g_{10}T, 0, T, 0)$, and the argument gives nothing. The conclusion can fail: for rule 150 with $w = 3$ and $M = 0$ the step $y[3]$ does not have $\mathrm{Hol} = K$, since the eventual image of $\Gamma_{3,0}$ is not strongly connected while that of $\Gamma_{2,0}$ is (Remark 5.8 of [1]): the tower of $\Gamma_{3,0}$ through the step $x[2]$ is that of $\Gamma_{2,0}$ followed by one memory step (the last part of Lemma 5.2, whose proof does not use $g_{11} = 1$), so the failure occurs at $y[3]$.

(3) The proof is not constructive. The walks ω and ω' exist because I_{d-1} is the whole group; they depend on M, and their bottom layers are arbitrary periodic input sequences rather than the layers of a tail.

Proposition 7.5. *Let $f \in \{30, 45, 75, 135\}$, $w \ge 1$, $M \ge 0$, and $-M + 1 \le d \le w$. Then* $\mathrm{Hol} = K$ *at the step $L_{\nu+1} = y[d]$.*

Proof. For 30 this is Lemma 4.13. For the others it follows from Proposition 6.1 for $d \le 1$. Let $d \ge 2$. For 75 and 45 we use Corollary 5.5 with the histories of this subsection: for 75, the wall; for 45, the period-4 loop for $d = 2$ and the restricted wall for $d \ge 3$. All of them are defined on all layers below their top, so they serve every M. For 135 we use Proposition 7.3. Its hypothesis (i) is Proposition 6.1. For (ii), the history below 2 given above for 135 has period 3, so by Lemma 5.2 it gives a closed walk of length 3 in G_2, for every $M \ge 0$ and $w \ge 2$. □

8. The lift certificate

This subsection is not used in any other proof. Proposition 7.3 gives full holonomy for 135 at every depth without exhibiting histories. The following finite computation produces explicit histories depth by depth. It confirms the behind-layer case of Proposition 7.5 by an independent route: for 30 and 45 at every depth, and for 135 at the depths $d \le 105$. Its parameters are f, $n_{\max}$, $T_{\max}$ and the largest depth D (Section 10).

Nodes. A node of depth $k \ge 0$ is a pair (X, V). Here X is a time series, taken up to rotation, of the four bits $(Z_t^a(k-1), Z_t^a(k), Z_t^b(k-1), Z_t^b(k))$ over its minimal period m, and $V \ge 0$ is an integer. The node is *realized* by a periodic history Z on layers $\le k$ (that is, below $k+1$) of minimal period T_Z if the top two layers of Z have joint series X, up to a time shift of Z, and V is the 2-adic valuation $v_2(T_Z)$. Then $m \mid T_Z$.

Start. $\mathcal{R}_0$ consists of one node of depth 0 for each F-orbit of a configuration c with $\sigma^n c = c$, $n \le n_{\max}$. It is realized by the comoving diagram of c with phase 0 in copy a and the same diagram with layer 0 complemented in copy b (Lemmas 5.4 and 5.3(4)). Here V is v_2 of the minimal comoving period of the diagram.

Test. The bits φ_t^c for the step $y[k+1]$ read only the top two layers, so they are m-periodic. Over the period T_Z, $\ell_c(Z) = (T_Z/m)\sum_{t<m}\varphi_t^c$, and T_Z/m is odd exactly when $v_2(m) = V$. The test $\mathrm{test}(\mathcal{R}_k)$ is true if some node of $\mathcal{R}_k$ has $v_2(m) = V$ and $\sum_{t<m}\varphi_t^a \ne \sum_{t<m}\varphi_t^b$. Then any realization Z of that node has $\ell_a(Z) \ne \ell_b(Z)$, and Corollary 5.5(b) gives $\mathrm{Hol} = K$ at the step $y[k+1]$, for every M.

Lift. By (Z1), a layer $k+1$ extending Z in copy c is a series with $Z_{t+1}(k+1) = Z_t(k+1) \oplus \varphi_t^c$. If $\sum_{t<m}\varphi_t^c = 0$ there are two such series, of period m, one for each initial value. If the sum is 1, there are again two such series, each of period $2m$ and not of period m; so the minimal period m' of either one divides $2m$ and does not divide m, which gives $v_2(m') = v_2(m) + 1$. (The minimal period need not be $2m$: for $m = 3$ the series 010101 has minimal period 2.) Only this valuation is used. In this case only the series with initial value 0 is kept. The new history has minimal period $\mathrm{lcm}(T_Z, m_a, m_b)$, where m_c is the minimal period of the new layer in copy c, and its 2-adic valuation is $\max(V, v_2(m_a), v_2(m_b))$. For each node of $\mathcal{R}_k$ and each choice of new layers, form the joint series of layers $(k, k+1)$; if its length is at most $T_{\max}$, it gives the node $(X', \max(V, v_2(m_a), v_2(m_b)))$. The set of these nodes is $\mathcal{R}_{k+1}$. Each node of $\mathcal{R}_{k+1}$ is realized, by extending a realization of its parent; condition (Z2) is unaffected since $k+1 \ge 1$.

By construction $\mathcal{R}_k$ is a *subset* of the nodes realized by periodic histories: lifts of period above $T_{\max}$ are omitted, and when both copies have odd increment sum, only one of the two relative phases of the new layers is included. This suffices, since only the existence of one good node per depth is used.

Why a repetition covers all depths. The new node set $\mathcal{R}_{k+1} = \mathcal{N}(\mathcal{R}_k)$ is computed by a map $\mathcal{N}$ that does not depend on k, and the test $\mathrm{test}(\mathcal{R}_k)$ depends only on $\mathcal{R}_k$. Hence if $\mathcal{R}_{k_0+\varrho} = \mathcal{R}_{k_0}$ for some $k_0 \ge 0$ and $\varrho \ge 1$, then $\mathcal{R}_{k+\varrho} = \mathcal{R}_k$ for all $k \ge k_0$, by induction, and $\{\mathcal{R}_k : k \ge 0\} = \{\mathcal{R}_0, \dots, \mathcal{R}_{k_0+\varrho-1}\}$. If $\mathrm{test}(\mathcal{R}_k)$ holds for $0 \le k < k_0 + \varrho$, it holds for all $k \ge 0$. Then $\mathrm{Hol} = K$ at every step $y[d]$ with $d \ge 1$, for every M.

Proposition 8.1 (computer-assisted)**.** (1) *For $f = 45$, $n_{\max} = 2$, $T_{\max} = 4$: $\mathrm{test}(\mathcal{R}_k)$ holds for $0 \le k \le 22$, and $\mathcal{R}_{23} = \mathcal{R}_{19}$ (here $k_0 = 19$, $\varrho = 4$, so the tests needed are those with $k \le 22$). Hence $\mathrm{Hol} = K$ at $y[d]$ for every $d \ge 1$.*

(2) *For $f = 30$, $n_{\max} = 2$, $T_{\max} = 1$: $\mathrm{test}(\mathcal{R}_k)$ holds for $0 \le k \le 2$, and $\mathcal{R}_3 = \mathcal{R}_2$.*

(3) *For $f = 135$, $n_{\max} = 4$, $T_{\max} = 12$: $\mathrm{test}(\mathcal{R}_k)$ holds for $0 \le k \le 104$, so $\mathrm{Hol} = K$ at $y[d]$ for $1 \le d \le 105$, and the sets $\mathcal{R}_k$, $0 \le k \le 105$, are pairwise distinct.*

This repeats the case $d \geq 1$ of Proposition 7.5 by an independent route, for 45 and 30 at every depth and for 135 at the depths $d \leq 105$.

Proposition 8.2. *For $f \in \{30, 45, 75, 135\}$ the conclusions of Theorem 3.1 hold for all $w \geq 1$ and $M \geq 0$.*

Proof. Parts (1), (2) and the size of $\mathcal{T}$ hold for every left-permutive f (Proposition 4.7 and Lemma 4.4), and part (3) holds once $\mathrm{Hol} = K$ at every step $y[d]$ with $-M + 1 \leq d \leq w$, by the proof of Theorem 3.1. This is Proposition 7.5. The consequences listed in Theorem 3.1 follow from (1)–(3) as shown after that theorem. □

9. The class of 135 behind the front

For 135 the holonomy behind the front is full at every depth (Proposition 7.5), and the conclusions of Theorem 3.1 hold for every w (Proposition 8.2). The proof differs from that for the other three rules in one respect: it gives no uniform family of histories (Remark 7.4(3)). By Corollary 5.5(b), full holonomy at depth d is witnessed by one periodic history below d with $\ell_a \neq \ell_b$. For 30 and 75 stationary histories do this at every depth $d \geq 2$. For 45 a history of period 4 does so at $d = 2$, and for $d \geq 3$ a wall of period 2 does so after a top flip (Corollary 5.5(c)). For 135 the certificate of Section 8 gives such histories at every depth $d \leq 105$, and the node sets $\mathcal{R}_0, \ldots, \mathcal{R}_{105}$ of Proposition 8.1(3) are pairwise distinct. We do not know whether rule 135 has a wall like that of 45, that is, a periodic history Z defined on all layers $i \in \mathbb{Z}$ with $N_a(Z, i) \neq N_b(Z, i)$ for all large i. Nothing in this paper depends on the answer.

10. Computations

The programs named below accompany [1], in its directory `code/`; the file `README.md` there states, for each program, what it computes and its expected output. Proposition 8.1 is the only statement of this note that is computer-assisted, and its consequences for the holonomy are also proved by hand (Proposition 7.5); the other computations confirm results proved by hand.

The lift certificate (Proposition 8.1). The program `lift_cert.py` computes the node sets $\mathcal{R}_k$ as specified in Section 8, with the parameters $(f, n_{\max}, T_{\max})$ of the proposition, up to the depth at which the repetition stated there occurs for (1) and (2), and up to depth 105 for (3). Each new set is compared with all the sets before it; for (3) the comparison is through SHA-256 digests of the sorted lists of nodes, and the 106 digests are pairwise different. The consequences of Proposition 8.1 for the holonomy are also proved by hand (Proposition 7.5), and no other proof depends on this computation.

The two-copy window graph (Theorem 3.1 and Proposition 8.2). For $f = 30$ and all (w, M) with $w+M \leq 6$, and for $f = 45, 75, 135$ and $(w, M) \in \{(1,0), (2,0), (3,0), (4,0), (1,1), (2,1), (3,1), (1,2)\}$, the graph $\Gamma_{w,M}$ built from the cell rule has an eventual image of $2^{3w+2}4^M$ states, which is strongly connected and is reached by every walk of length $\lceil w/2 \rceil$.

The holonomy behind the front (Lemma 7.2 and Proposition 7.3). For 135 the graphs $\Gamma_{w,M}$ are built directly from the cell rule for $2 \leq w \leq 7$ with $M = 0$, and for $(w, M) = (2,1)$, $(3,1)$, $(4,1)$, $(5,1)$, $(6,1)$, $(2,2)$, $(3,2)$, $(4,2)$, $(2,3)$. In every case the eventual image has $2^{3w+2}4^M$ states, the graph G_w at the step $y[w]$, obtained from it by forgetting the layer $y[w]$, is strongly connected, and the values $\Psi_w(\varsigma)$ of its closed walks are all 32 elements of $(\mathbb{Z}/2)^5$. The same holds for the other seven non-affine left-permutive rules for $2 \leq w \leq 4$, $M = 0$. For the affine rules 90, 105 and 150 with $2 \leq w \leq 4$, $M = 0$, the values of all closed walks generate a group of order at most 8 (order 8 for 90, whose graph is not strongly connected; order 2 or 4 for 105 and 150). The supplementary material contains four programs for this, which use the cell rule only and neither the layer equation nor the conditions (C_j). The program `ref_check.py` enumerates

all window states, iterates to the eventual image, and finds the exact set of values Ψ_w of the closed walks at a vertex by a search in $G_w \times (\mathbb{Z}/2)^5$; with the option `flip` it tests that the top flips and the swap of Lemma 7.2 are automorphisms of G_w that change ψ_w as stated there. The program `ref_fast.py` builds the eventual image by a breadth-first search from one of its states, counts the strongly connected components, and computes the group generated by the values Ψ_w from the potentials of a spanning tree; it is used for the larger cases $w = 6, 7$ with $M = 0$ and $(w, M) = (5, 1)$, $(6, 1)$, $(4, 2)$. The program `ref_affine.py` does the same over all components for the affine rules, and `ref_base.py` checks that the history below 2 of Section 7 for 135 is a closed walk of length 3 in G_2 with $\Psi_2 = (1, 0, 1, 1, 1)$ for $M \leq 3$, and that sampled closed walks of G_{d-1} with $\ell = (0, 0)$ lift to closed walks of G_d of the same length.

Use of AI tools. Large language models (Anthropic's Claude models) were used extensively in this work: to propose and develop proofs, to design, write and run the computer searches and their independent re-implementations, to check proofs and computations, and to draft the manuscript. The models are not authors. The author has reviewed the content and takes full responsibility for it. All code and computational output accompany [1] (Section 10).

References

[1] M. Dayah, Fronts at the speed of light: rigidity of permutive second-order cellular automata, in preparation.

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